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Assignment 1

Q1>

Ans:: Source :- Source is where the data is organized. Typically its a computer, but it can be any othe electric equipment which can generate data for transmissio to some destination. Data sent is represented as $x(t)$

Transmitter :- As data cannot be sent in its native form its necessary to convert it to signal. This is performed with help of modem. The signal that is sent by transmitter is represented by $s(t)$.

Communication Medium :- Signal can be sent to receiver through a communication medium, which could be a simple twisted pair of wire, a coaxial cable, optic fibre or wireless communication system. Signal that comes out of communication medium is $s'(t)$ which is different from $s(t)$ that was sent by transmitter. Its due to various impairments that signal suffers as it passes through communicati medium. Eg (noise)

Receiver :: Receiver receives signal $s'(t)$ and converts it to dat $d'(t)$ before forwarding to destination. Destination which receives data is not $d(t)$ due to data corruption.

Destination :- Destination is where data is absorbed. Again it can be computer system and so on.

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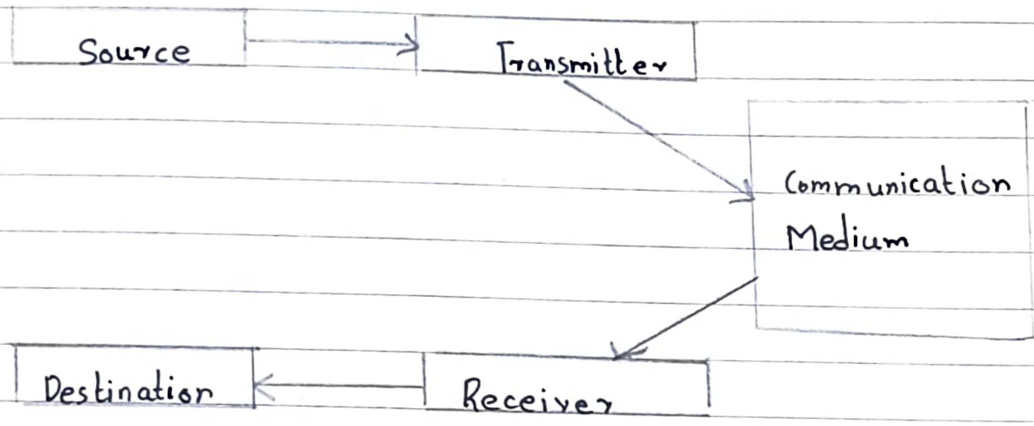

Q2>

Ans:: Data:: Data refers to information that conveys some meaning based on some mutually agreed up rules or conventions between a sender and reciever and today it comes in a variety of forms such as text, graphics, audio, video and animations.

Signal:: Its a electrical or optical representation of data, which can be sent over a communication medium. Stated in mathematical terms, a signal is merely a function of data. Eg:: Microphones convert voice data into voice signal which can be sent over a pair of wire. Analog signals are continuous valued where as digital signals are discrete valued. Independent variables of signal could be time, space, or integers.

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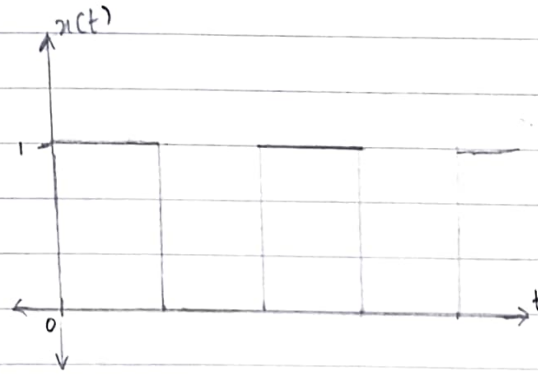
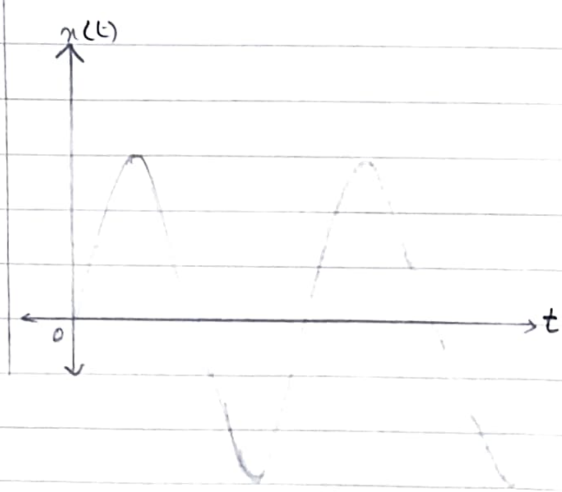
Q3>

Digital Signal	Analog Signal
→ Discrete signal values represents only finite values (0, 1).	Continuous signal values represents an infinite number of values within a range.
→ Has information represented by ~~continuously varying physical~~ discrete voltage or logic leves	Has information represented by continuously varying physical quantities
→ Less susceptible to noise due to fixed threshold levels	Highly susceptible to noise and interference
→ Can be regenerated after distortion by accurately using repeaters.	Cannot be perfectly regenerated once distorted.
→ Signal quality is maintained over long distances	Signal quality degrades as distance keeps increasing.
→ Error detection and correction techniques are possible	Error detection and corrections are difficult to ~~imp~~ implement.

→





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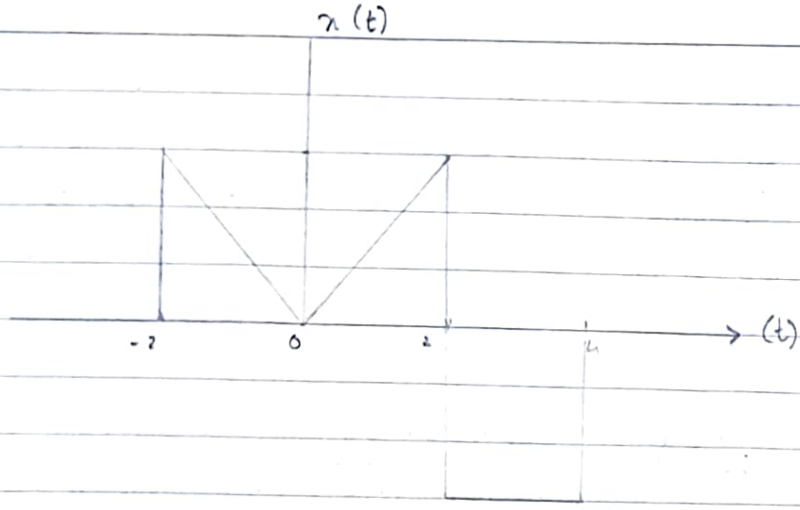
Q1)



Sketch i) $x(2t+3)$ ii) $x(-t-4)$

*Note Priority F . SC SH

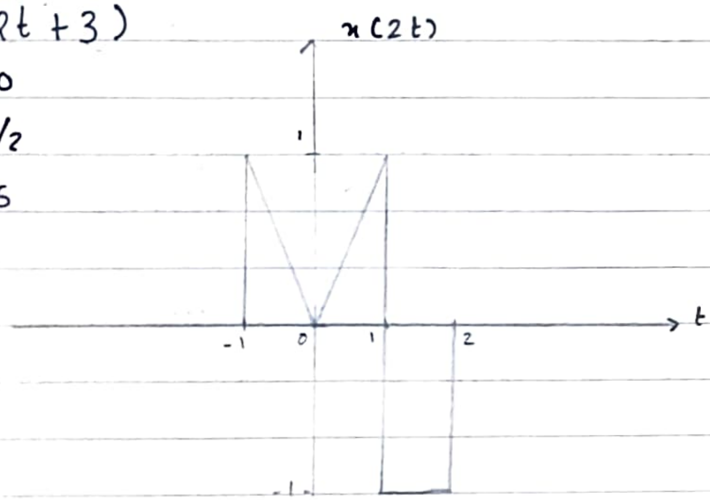
While performing shift, perform the = 0 rule on whole equation

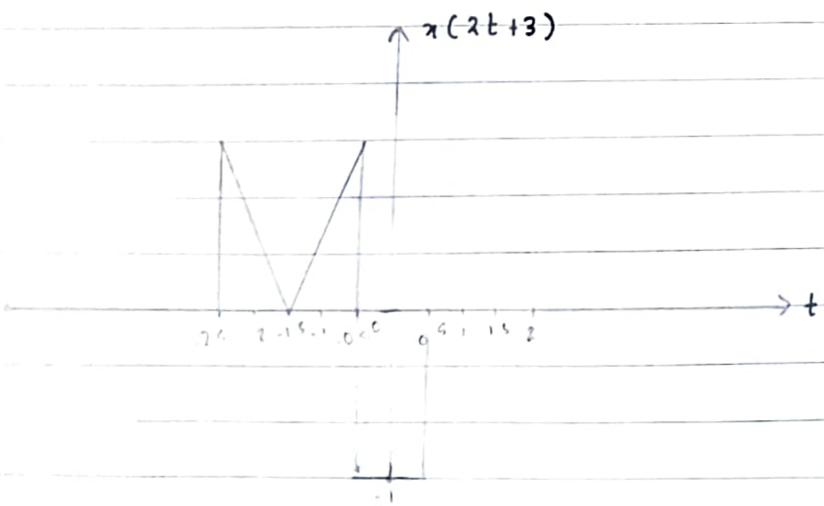
i) $x(2t+3)$

$2t+3=0$

$t = -3/2$

$t = -1.5$





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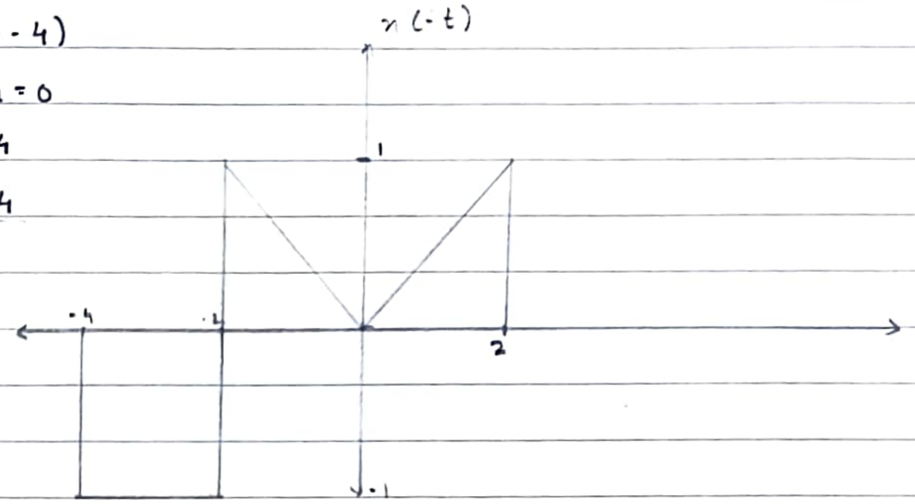
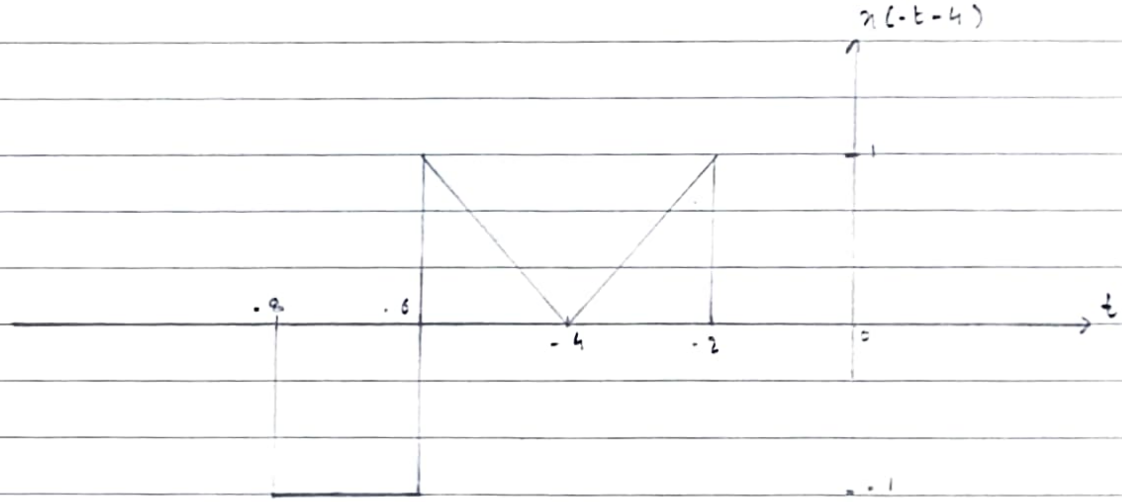
ii) $x(-t-4)$

$-t-4=0$

$-t=4$

$t=-4$

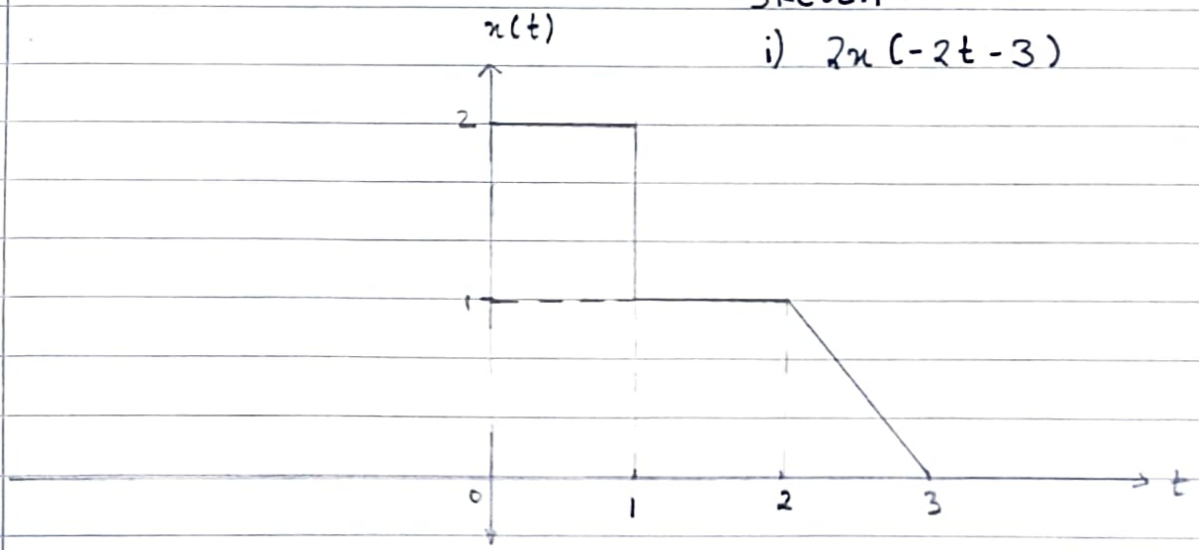




Q2)

Sketch :

i) $2x(-2t-3)$



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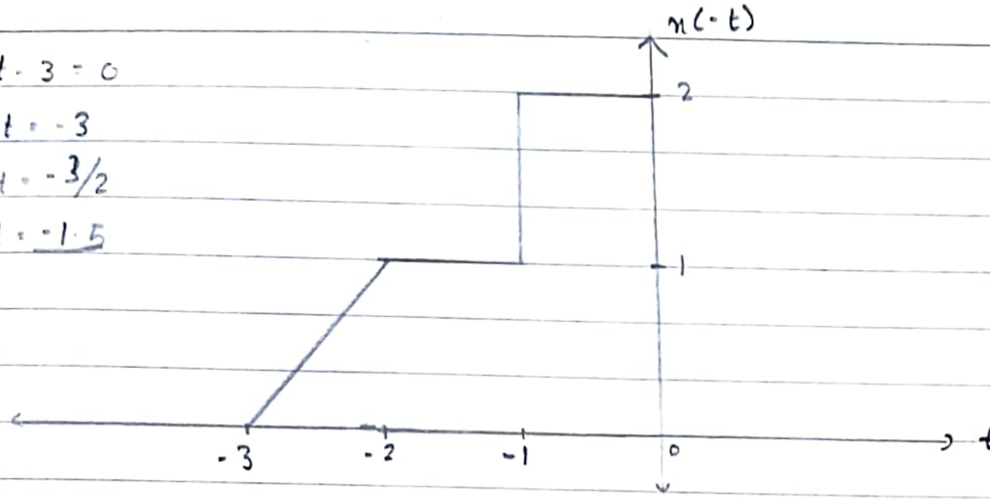
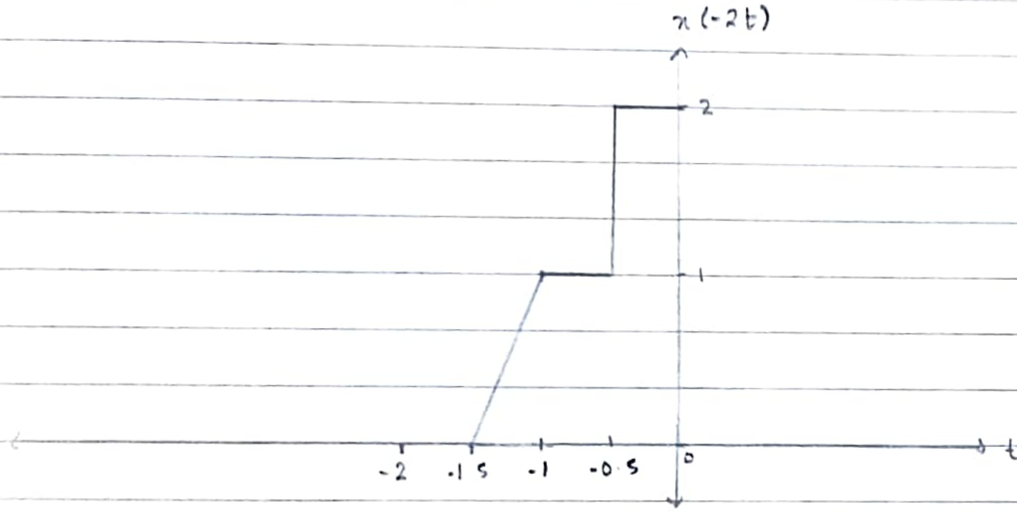
$-2t-3=0$

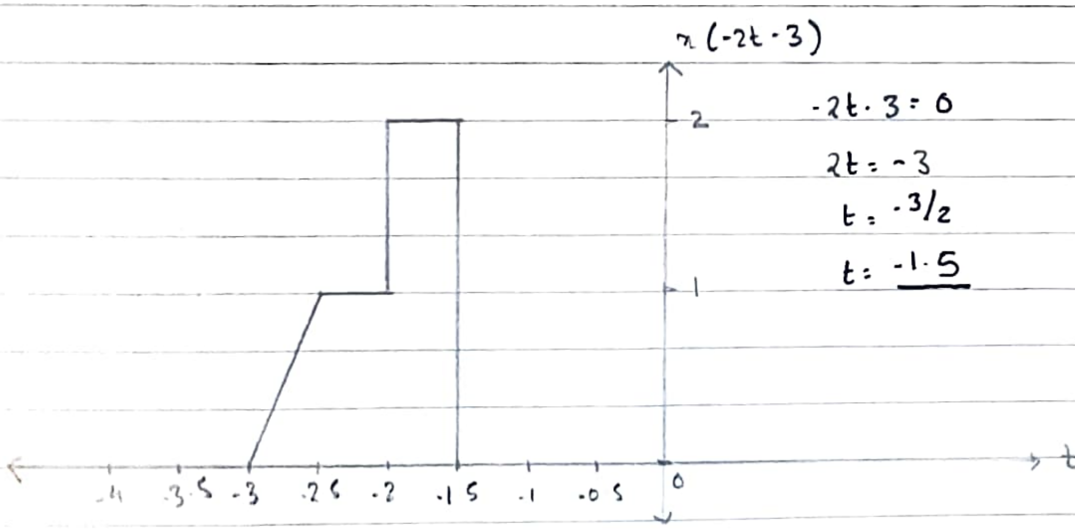
$2t=-3$

$t=-3/2$

$t=\underline{-1.5}$







$-2t-3=0$

$2t=-3$

$t=-3/2$

$t=\underline{-1.5}$

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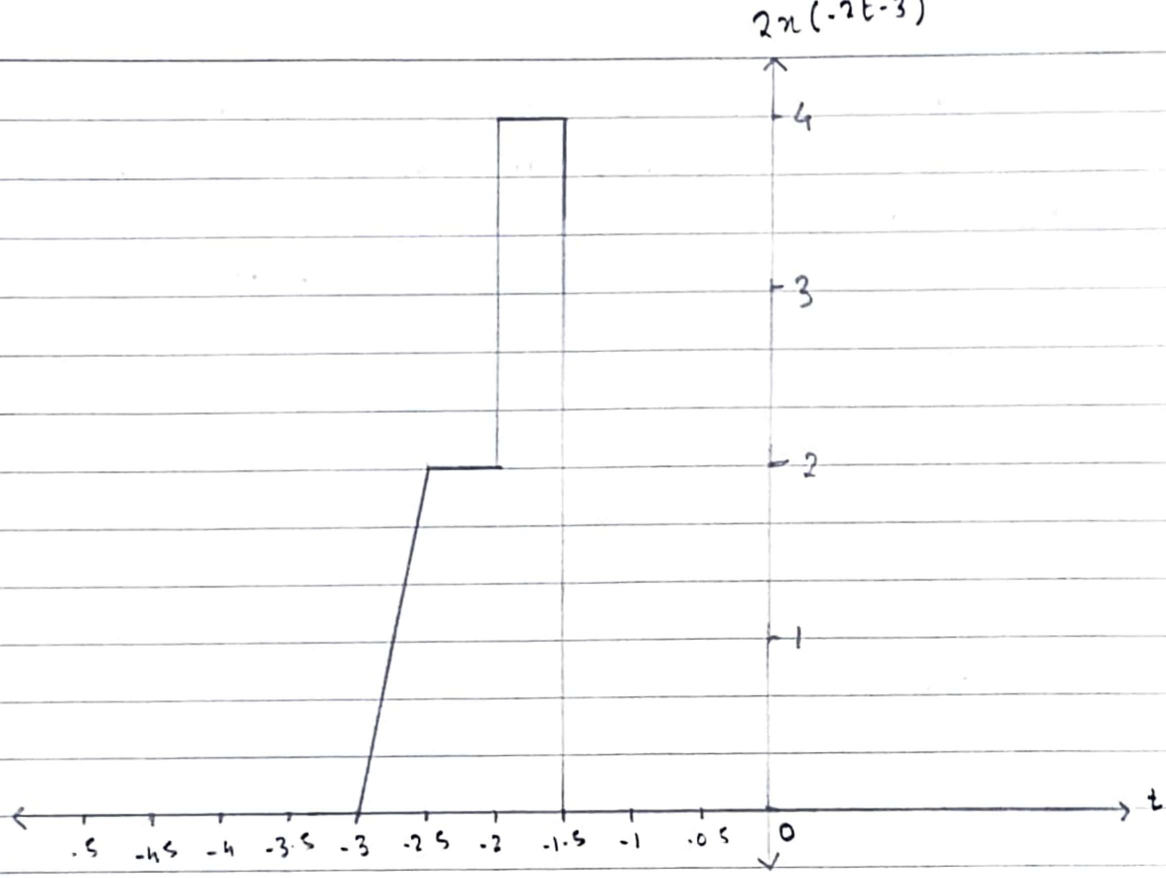
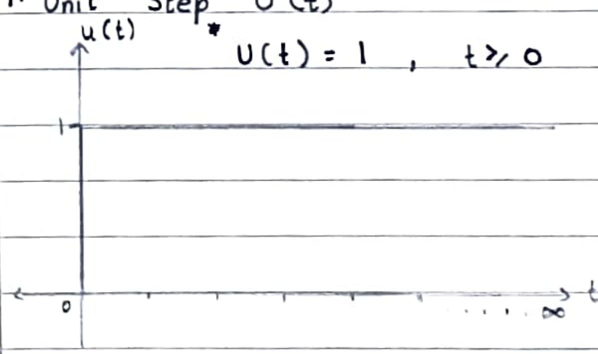


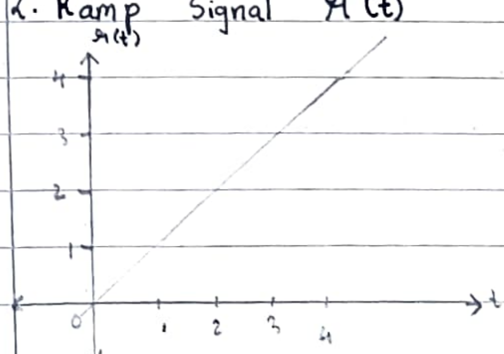
Standard Continuous Time Signal.

1. Unit Step $U(t)$

$U(t) = 1$, $t \geq 0$; $U(t) = 0$, otherwise conditions



2. Ramp Signal $r(t)$



$r(t) = t$ $t \geq 0$

$= 0$ otherwise

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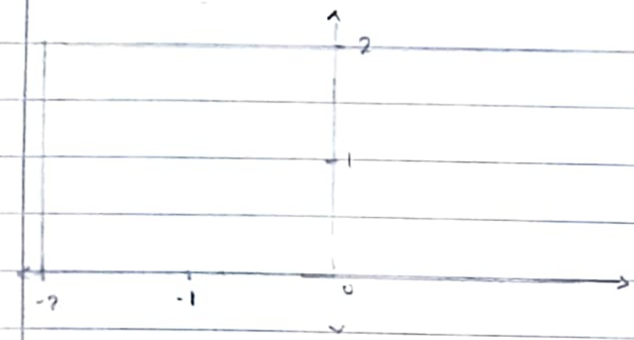
Construct a CTS for given equation

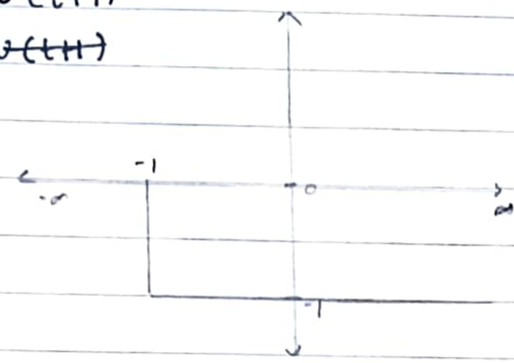
$x(t) = 2u(t+2) - u(t+1) - r(t-1) + r(t-2)$

Ans: $2u(t+2)$

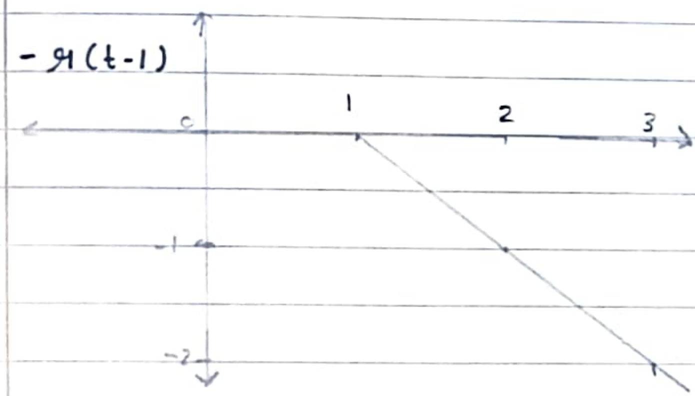


$-u(t+1)$

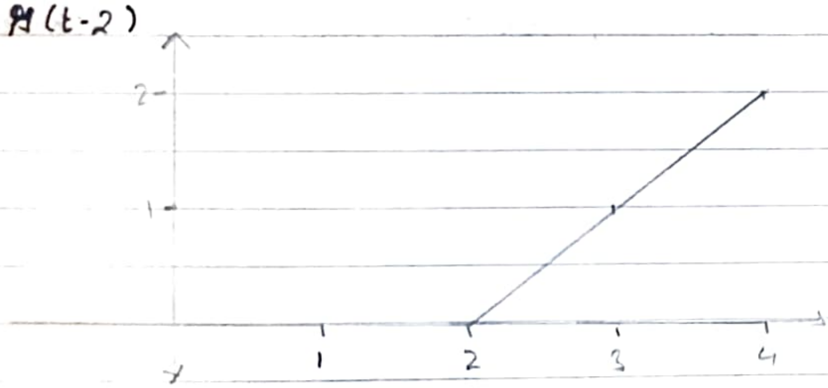
~~$u(t+1)$~~



$-r(t-1)$



$r(t-2)$



$x(-2) = 2u(0) - u(-1) - r(-3) + r(-4) = 2(1) - 0(-0 - 0 = 2$

$x(-1) = 2u(1) - u(0) - r(-2) + r(-3) = 2(1) - 1 - 0 - 0 = 1$

$x(0) = 2u(2) - u(1) - r(1) + r(-2 = 2(1) - (1) - 0 - 0 = 1$

$x(1) = \quad = 2(1) - (1) - (0) - 0 = 1$

$x(2) = \quad = 2(1) - (1) - (1) - 0 = 0$

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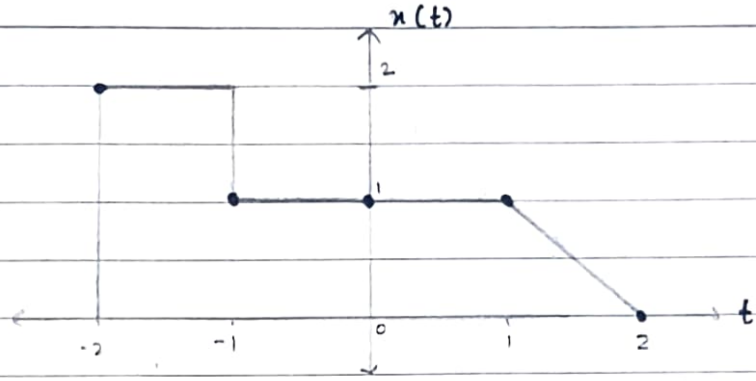
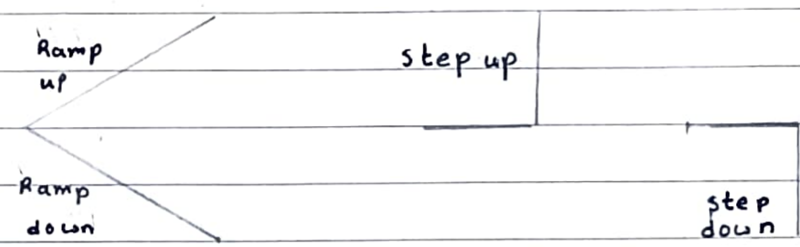
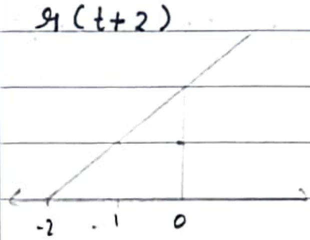




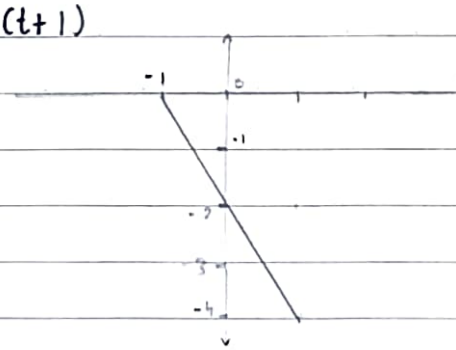
Always see nature of destination just not for starting.

$$x(t) = r(t+2) - 2r(t+1) + r(t-1) + 2u(t-3) - u(t-4)$$

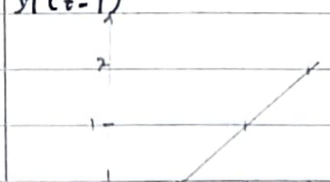
$r(t+2)$



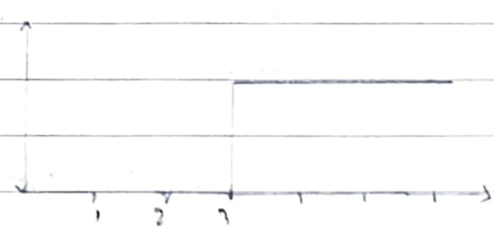
$-2r(t+1)$



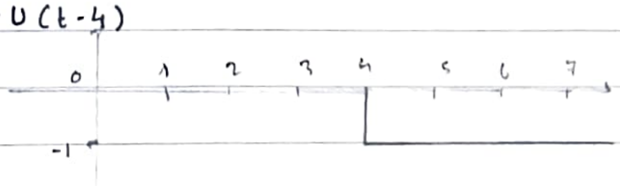
$r(t-1)$



$2u(t-3)$



$-u(t-4)$



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$$x(-2) = r(0) - 2r(-1) + r(-3) + 2u(-5) - u(-6) = 0 - 0 + 0 + 0 + 0 - 0 = 0$$

$$x(-1) = r(1) - 2r(0) + r(-2) + 2u(-4) - u(-5) = 1 - 0 + 0 + 0 + 0 - 0 = 1$$

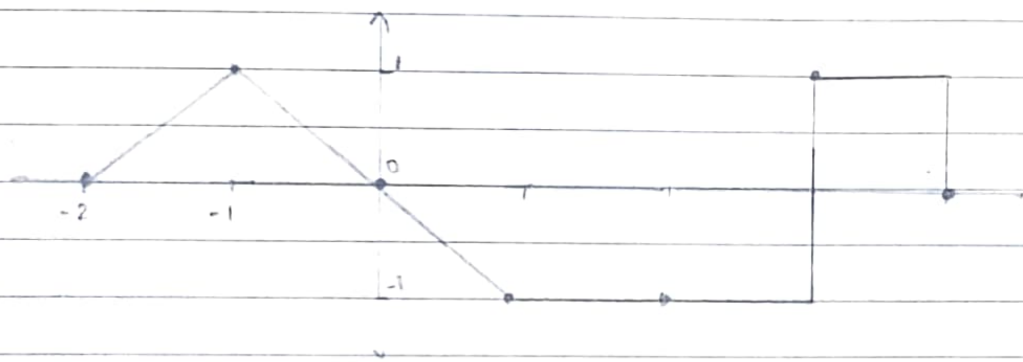
$$x(0) = r(2) - 2r(1) + r(-1) + 2u(-3) - u(-4) = 2 - 2 + 0 + 0 + 0 - 0 = 0$$

$$x(1) = r(3) - 2r(2) + r(0) + 2u(-2) - u(-3) = 3 - 4 + 0 + 0 + 0 - 0 = -1$$

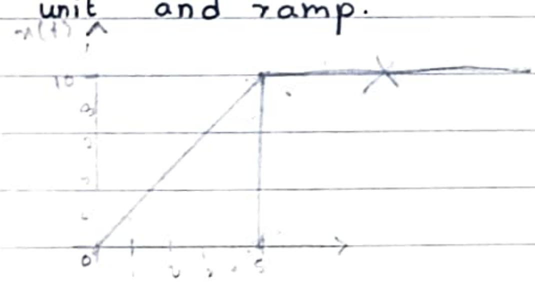
$$x(2) = r(4) - 2r(3) + r(1) + 2u(-1) - u(-2) = 4 - 6 + 1 + 0 + 0 - 0 = -1$$

$$x(3) = r(5) - 2r(4) + r(2) + 2u(0) - u(-1) = 5 - \overset{8}{10} + 2 + 2 - 0 = 1$$

$$x(4) = r(6) - 2r(5) + r(3) + 2u(1) - u(0) = 6 - 10 + 3 + 2 - 1 = 0$$



From signal obtain the equation of the signal in terms of unit and ramp.



* Note to nullify ramp signal equal and opposite value sign.

$$x(t) = 2r(t) - 2r(t-5) - 10u(t-5)$$

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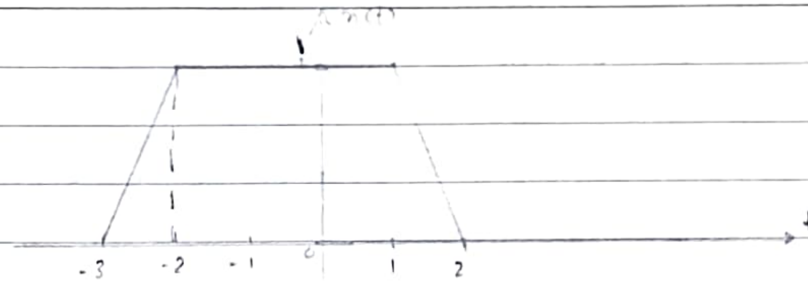
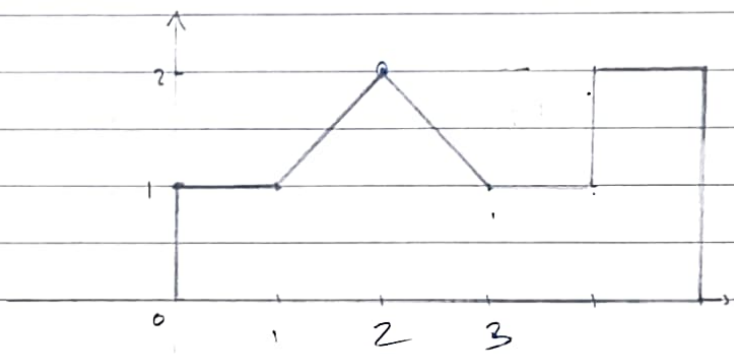


$x(t) = r(t+3) - r(t+2) - r(t-1) + r(t-2)$



$x(t) = u(t) + r(t-1) - r(t-2) - r(t-2) + r(t-3) + \cancel{2u(t)} \cancel{2u(t)}$

$+ u(t-4) - 2u(t-5)$

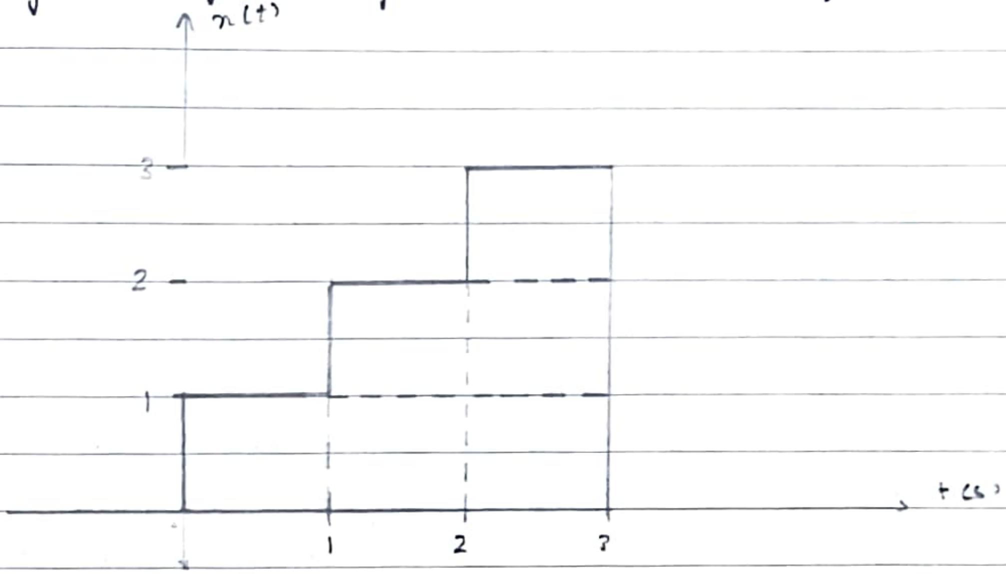
$= u(t) + r(t-1) - 2r(t-2) + r(t-3)$

$+ u(t-4) - 2u(t-5)$

A1

From given signal diagram deduce its equation.

Q1>



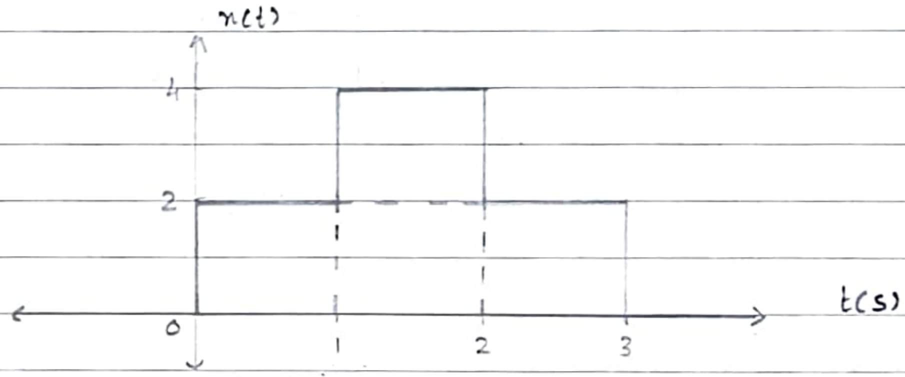
$x(t) = u(t) + u(t-1) + u(t-2) - 3u(t-3)$

$x(2) = u(2) + u(2-1) + u(2-2) - 3u(t-3) \Rightarrow 2 = 1 + 1 + 0 + 0 \therefore 2 = 2$

$x(0) = u(0) + u(0-1) + u(0-2) - 3u(0-3) \Rightarrow 1 = 1 + 0 + 0 + 0 \therefore 1 = 1$

$\therefore$ Equation upholds to be true.

Q2>



$x(t) = 2u(t) + 2u(t-1) - 2u(t-2) - 2u(t-3)$

$x(1) = 2u(1) + 2u(0) - 2u(1-2) - 2u(1-2) \Rightarrow 4 = 2+2+0+0 \therefore 4 = 4$

$x(2) = 2u(2) + 2u(2-1) - 2u(2-2) - 2u(2-3) \Rightarrow 2 = 2+2-2+0 \therefore 2 = 2$

$\therefore$ Equation upholds to be true.

A1

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Basic Signal Operations:

> Time Shifting: (Time-delayed signals & Time-advanced signals)
Time shifting is an important operation that is used in many signal-processing applications. For eg: a time delayed version of signal is used when performing autocorrelation. example of time shifting in autocorrelation is finding a repeating beat in a noisy audio recording. Autocorrelation ~~is~~ involves comparing a signal to a "lagged" copy of itself to find patterns. (Time Delay Neural Networks)

- Orignal Signal: A recording of a drum beating every 1s but, with a lot of static noise.
- Time Shifted Signal: Second copy of that same recording which you manually slide forward in time.

> Time Scalinging:
When time scaling is performed we change the rate at which the signal is sampled. Changing the sampling rate of a signal is employed in the field of speech processing. A paticular example is time scaling algorithm based system developed to read text to visually impaired.

> Time Reversal: (Folding)
Time reversal is an important preliminary step when computing the convolution of signals: one signal is kept in its orignal state while other is mirror image and slid along the former signal to obtain result. Hence time reversal is used in edge detection. Time reversal technique in form of time reverse numerical simulation (TRNS) is used to determine defects. TRNS aids in finding out exact position of a notch which is a part of structure along with guided wave propagates.